

Project Portfolio CPS

- Project Portfolio CPS..... 1**
 - Week 1:.....2
 - Week 2:.....2
 - Week 3:.....3
 - Week 4:.....4
 - Week 5:.....4
 - Week 6:.....4
 - Week 7:.....7
 - To be expanded:..... 10
 - Week 8..... 10
 - System boundary..... 11
 - Internal to the CPS: the sensing buoy, the telemetry link, and the computing stack (twin, estimator, decision loop)..... 11
 - External: the river flow, the unknown contamination plume, and physical constraints such as banks and obstacles..... 11
- Bibliography..... 12**

Process Overview

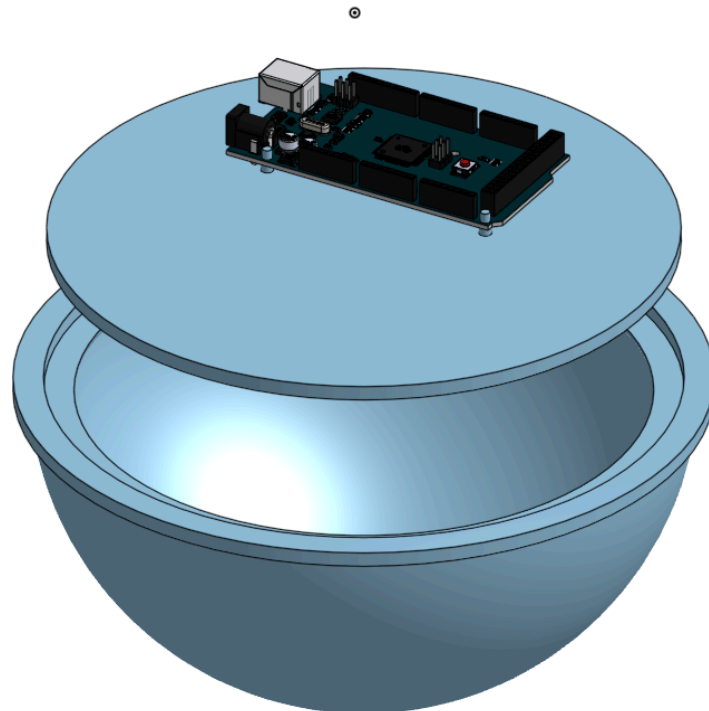
Week 1:

We discussed and decided to start the work by testing the sensors first. The purpose is to calibrate them and perform some tests in the lab with pollutants like saline solutions, etc. Then we analyze the data we receive and build the digital twin that would receive the data and raise a flag if the perceived values are higher than the set threshold.

To begin with the project, we ordered a [starter kit](#) that included a pH sensor and a conductivity sensor, along with an [Arduino Mega 2560](#), a [Dissolved Oxygen sensor](#), and an [analog signal isolator](#).

Week 2:

The items have been ordered but haven't arrived yet. We are writing some basic code so we have a starting point for when the parts arrive, we are also looking into the practical usage of different hardware components and how we can use the retrieved data in our digital twin. We are also creating a few examples of the 3D model for the buoy with the prediction we have made in the first semester.



This week's progress is mostly based on research and looking at similar projects online, in order to identify their drawbacks and limitations, which we can use to further shape our ideas and improve our work.

Week 3:

We make a work plan for the following weeks:

Week 3	Make work plan
Week 4	Test the sensors
Week 5	Integrate sensor data to digital twin & discuss communication tech & Order parts
Week 6	Test communication & Discuss power & Order parts
Week 7	Test power & Discuss the shell & Order motors
Week 8	Make the shell and put everything together
Week 9	----- Holiday
Week 10	----- Holiday
Week 11	Test Motors
Week 12	Test buoy

Week 4:

No new updates; sensors haven't arrived yet and we still have nothing to work with in practice.

Week 5:

The sensors, the Arduino, and the communication modules have arrived. We calibrated the sensors with the respective buffer solutions that arrived with them by writing an Arduino script that includes reading the sensors and printing the values.

The next step was to test the communication model since we have two alternatives: two 4G/LTE shields with antennas and GNSS/GPS sensor. We tested only one of them and found a solution about the IoT cloud service. The Thingsboard platform, which is the chosen IoT interface, should enable communication with the sensors (water buoy) and the digital twin later on. The first shield with the two antennas (GNSS and 4G/LTE) worked quite well.

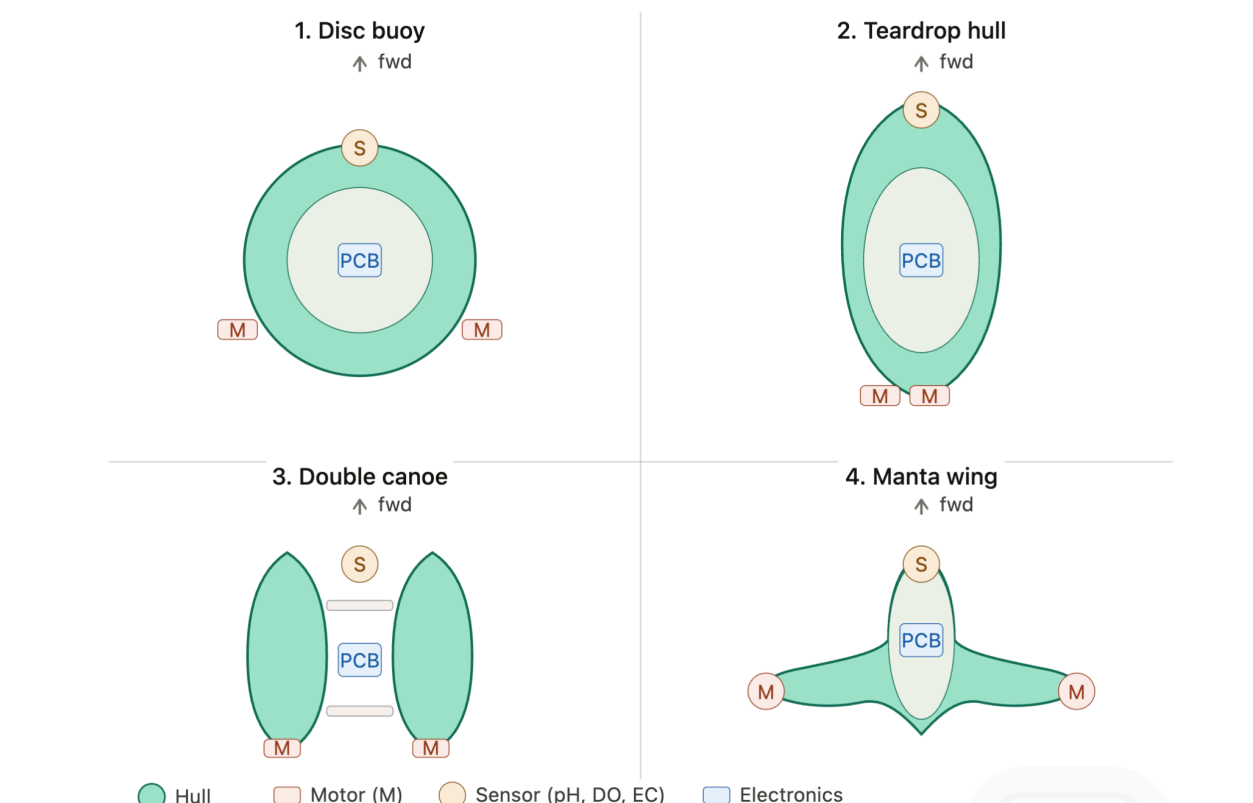
However, there is an issue with the dissolved oxygen sensor. Apparently, the sensor needs a solution to fill in the membrane cap before using it in the water.

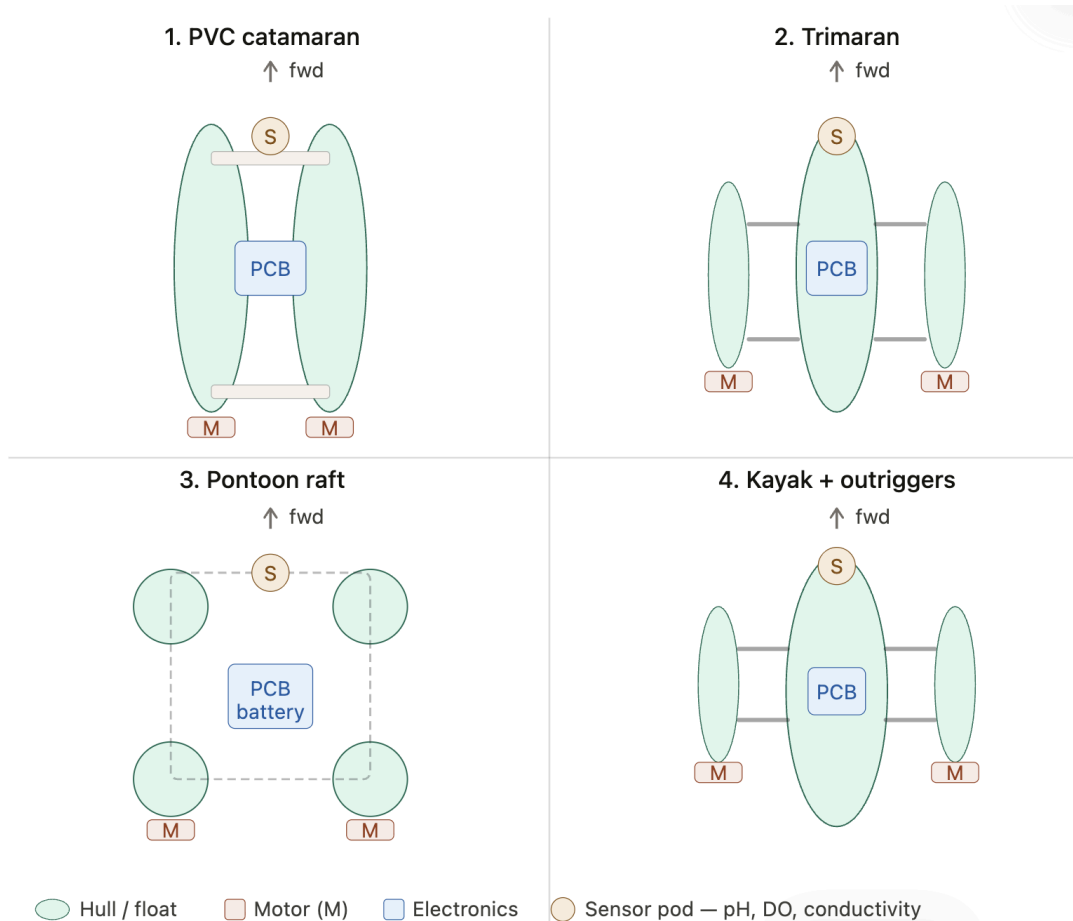
Week 6:

We found the solution for the Dissolved Oxygen sensor so that it can correctly read values from different solutions. We tested the GNSS sensor together with one shield; however, the GPS signal took forever to arrive, so we decided to go with the other alternative that was tested last week. We were able to set up the dashboard with different graphs and visualization tools so that it is easier to see what the sensors are reading and what data is being transmitted over 4G network.

We build a digital twin for the telemetry received from the sensor. We still need to check the threshold values and make sure the digital twin alerts the user when pollution is detected. This way, we can continue with the respective steps.

Regarding the structure of the buoy, we made the following comparison between the structures we could think of:





Our initial plan was to make the water buoy in the shape of a disk buoy. However, after a trade off analysis with the help of AI, we found that the Teardrop buoy would be a better solution if we made it look more like an RC boat.

Design	Roll stability	Capsize resist.	Current handling	Hydro-dynamics	Build simplicity	Sensor mounting	Electronics shelter	Best for
Disc buoy	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Slow/calm water, stationary sampling
Teardrop hull	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	River transit, long missions
Double canoe	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Balanced structure
Manta wing	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Showcase, low current
Catamaran	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Best first build, slow rivers
Trimaran	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Moderate current, neat build
Pontoon raft	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Calm water, heavy payload
Kayak & outriggers	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	● ● ● ● ●	Strong current, long range

The digital twin needs some further improvements, like backtracking, threshold values for alert detection, etc. Project can be accessed here: https://github.com/KbeFr/cps_bouy_dt

Week 7:

After careful consideration, we decided to purchase an RC boat, particularly on of those used for baits in fishing, as they have a big body and we can easily place our components inside. This would save us a lot of time trying to 3D print the whole structure, and going through the trial and error in a limited time.

We consulted [1-4], including the research and guidelines for the pH, conductivity, and dissolved oxygen values for the Meuse River. Here is the table of values according to the sources:

pH Thresholds

Zone	pH range	Meaning for Meuse
Critical	< 6.0	Acid spill, mine drainage, industrial discharge. Fish cannot survive below 6. Lethal within hours.
Warning	6.0 – 6.5	Below WFD good status. Possible acid rain/runoff.
Normal	6.5 – 8.5	WFD good ecological status. Meuse nominal: 7.5 to 8.2. All aquatic life is supported.
Warning	8.5 – 9.0	Algae bloom / eutrophication event. High photosynthesis consuming CO ₂ . Values reached during Meuse 2003 drought peak.
Critical	> 9.0	Severe alkaline discharge (cement, lime, industrial effluent). Fatal to most aquatic organisms.

EC Thresholds

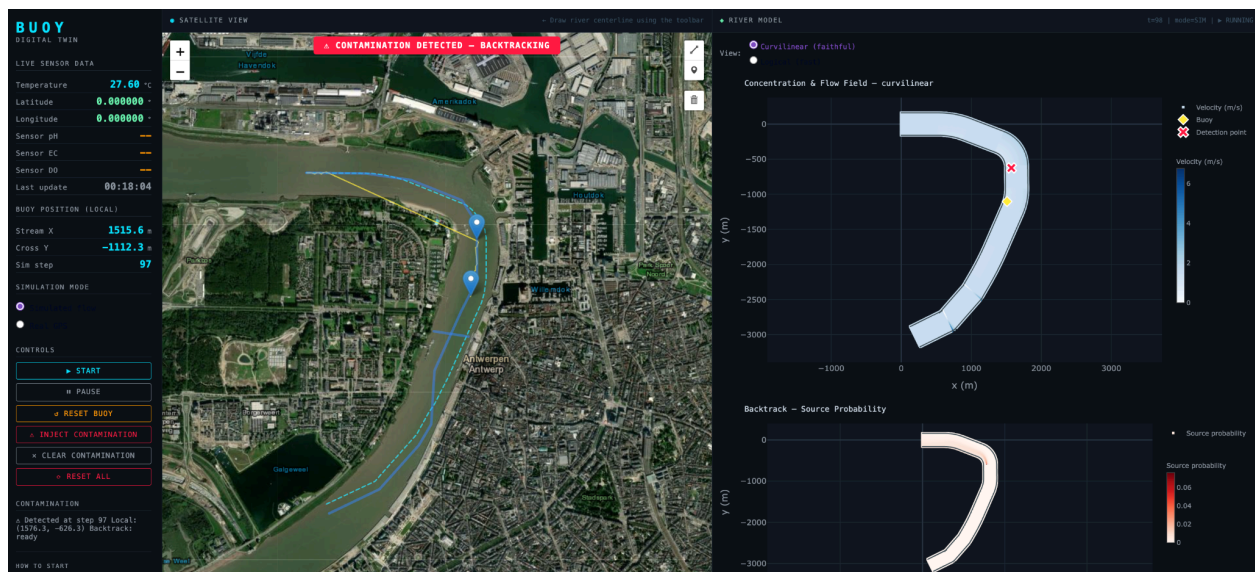
Zone	EC range	Meaning for Meuse
------	----------	-------------------

Very clean	< 200 $\mu\text{S/cm}$	Unusually clean/upstream mountain water. Atypical for the Belgian Meuse, which carries dissolved salts from Walloon limestone geology.
Healthy	200 – 700 $\mu\text{S/cm}$	Normal Meuse range. Limestone geology contributes natural calcium/bicarbonate ions. Aquatic life is fully supported.
Elevated	700 – 1000 $\mu\text{S/cm}$	Possible low-flow concentration, agricultural runoff (fertilisers), or urban wastewater dilution failure. Meuse reaches this during summer drought.
Polluted	> 1000 $\mu\text{S/cm}$	Strong indication of industrial discharge (chemical plant, mining), road-salt runoff, or sewage overflow. Chronic stress for sensitive invertebrates and fish.
Severe	> 2000 $\mu\text{S/cm}$	Major industrial or saline spill. Fish kill likely within 24 to 48 hrs. Lethal to freshwater species.

DO Thresholds

Zone	DO range	Meaning for Meuse
Super-saturated	> 12 mg/L	Algae bloom producing excess O_2 by day. Meuse 2003 drought: daytime supersaturation observed. Indicates eutrophication in progress and will likely flip to hypoxia at night.
Excellent	9 – 12 mg/L	Winter / cool season Meuse values.
Good	7 – 9 mg/L	Nominal Meuse summer values at cool temperature. WFD good status maintained. All fish supported.

Moderate	5 – 7 mg/L	Below WFD good status threshold (6 mg/L). Sensitive species stressed. Consistent with Meuse drought summers. Organic pollution possible.
Poor	3 – 5 mg/L	Hypoxic - Organic/sewage pollution likely. Immediate ecological damage.
Lethal	< 3 mg/L	Anoxic zone. Biochemical oxygen demand from heavy organic load consumes all O ₂ . Emergency-level event.



To be decided next week:

- 1- Boat purchase for the structure,
- 2- IMU sensor for heading
- 3- Energy support (battery)
- 4- Digital Twin features
- 5- Control with Arduino

Links for the boats:

[Boat 1](#)

[Boat 2](#)

[Boat 3](#)

[Boat 4](#)

[Boat 5](#)

[Boat 6](#)

Week 8

We decided to purchase Boat 4 from the links above, as it allows us to place the sensors in the water without interfering with the boat's overall structure. While the boat was on the way, we went to Campus Drie Eiken to fill in the cap of the sensor with the NaOH solution as required by the sensor documentation. Then we were able to test the sensors and integrate them through the platform of Thingsboard, while also setting the alerts and threshold detections in the platform as well.

Week 9 & 10 Holiday

/

Week 11

The boat arrived.

Tried to integrate our Arduino with the PCB of the boat, but it didn't work as the PCB is custom-made. We tried to reverse engineer it with pins, but then the microcontroller would not be compatible with our built circuit and scripts. We also decided to move everything to the ESP32 MC since we don't have that much space in the boat itself, and also, there is no need to have two microcontrollers that can serve the same purpose. So another decision was to keep the control of the boat as it is; we won't interfere with the structure for now, and we will keep our circuit on the outside.

Week 12

Next thing we did was measure the boat to provide a support and insulating structure that will be printed through a 3D print. This will keep the sensors intact while still keeping the boards and components waterproof. The purpose is to make minimal edits to the overall structure of the boat, while making sure that it provides the necessary support. Then we decided to make a custom PCB to connect all of the components with the ESP32 microcontroller. To make the sensor cables and boards neater, we bolted them to a small wooden board, and then the rest of the components would be connected through a PCB.

Week 13

We worked on the PCB, soldered everything, and tested the components. We decided that it would be nice if we added an IMU sensor to it, as it will help with the direction of heading for the digital twin, as well as the control of the boat through the digital twin commands as a future work. The integration of the IMU sensor to the whole circuit introduced an extra challenge for the pins that needed to be soldered.

Digital Twin improvements and feature refinements.

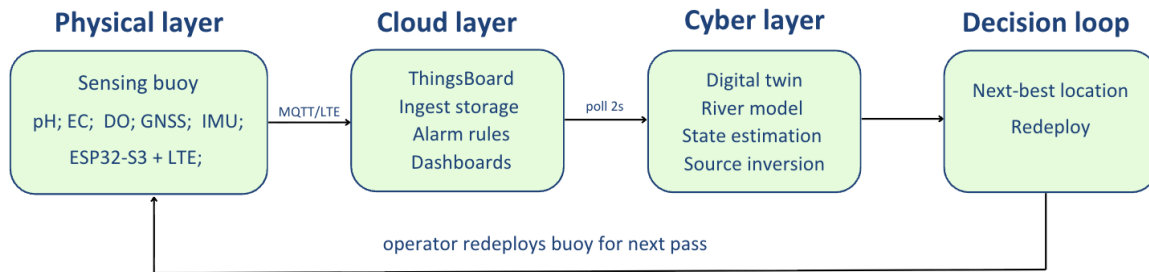
After the insulating material of the boat is dried out, we put everything into place and test the PCB with the battery and the new Arduino code that include all the sensors. We store the data in a .csv file through an SD Card. The data is important to help improve the digital twin even further.

Week 14

This stage was mainly to validate and test the boat in a real pond while making use of the GNSS coordinates, Thingsboard alarms and thresholds, data display, as well as the real measurements in an external environment. As expected the GNSS was not the most accurate, as the pond was located in a park with a lot of trees, however, the results were pretty good.

Systems Architecture

The platform is organized as four cooperating layers. The physical layer (the buoy) senses and transmits, the cloud layer brokers and stores, the cyber layer (digital twin) interprets and estimates, and a decision loop closes back to the operator, who can redeploy the buoy for the next measurement pass.

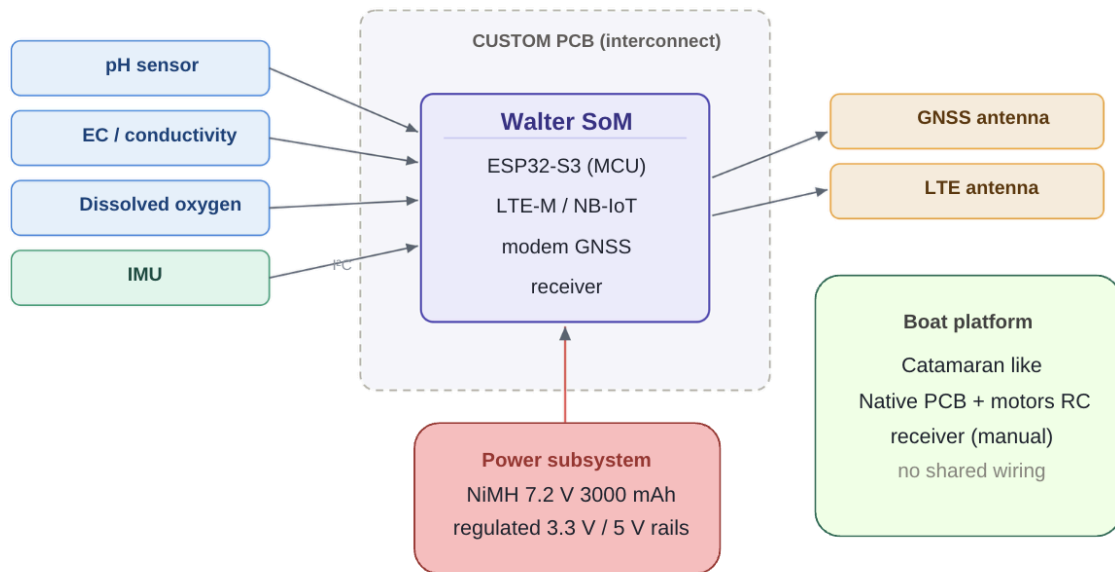


System boundary

Internal to the CPS: the sensing buoy, the telemetry link, and the computing stack (twin, estimator, decision loop).

External: the river flow, the unknown contamination plume, and physical constraints such as banks and obstacles.

On-board hardware architecture



Bibliography

- [1] M. T. H. Van Vliet and J. J. G. Zwolsman, "Impact of summer droughts on the water quality of the Meuse River," *Journal of Hydrology*, vol. 353, no. 1–2, pp. 1–17, May 2008. doi: 10.1016/j.jhydrol.2008.01.001
- [2] A. Catteau *et al.*, "Water quality of the Meuse watershed: Assessment using a multi-biomarker approach with caged three-spined stickleback (*Gasterosteus aculeatus* L.)," *Ecotoxicology and Environmental Safety*, vol. 208, p. 111407, Jan. 2021. doi: 10.1016/j.ecoenv.2020.111407
- [3] RIWA-Maas, "Drinking water relevant substances in the River Meuse: Evaluation and update of the substance lists 2021," RIWA-Maas Report, 2021. [Online]. Available: <https://www.riwa-maas.org/en/publicatie/1892/>
- [4] European Parliament and Council of the European Union, "Directive 2000/60/EC of the European Parliament and of the Council establishing a framework for Community action in the field of water policy," *Official Journal of the European Communities*, L 327, pp. 1–73, Dec. 2000. [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32000L0060>